

CSCI 467: Foundations of Machine Learning

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USC CSCI 467, Fall 2026
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Today's Plan

- What is machine learning?
- Where is machine learning used?
- Course Logistics
- Bird's Eye View of the Schedule
- Start on linear regression

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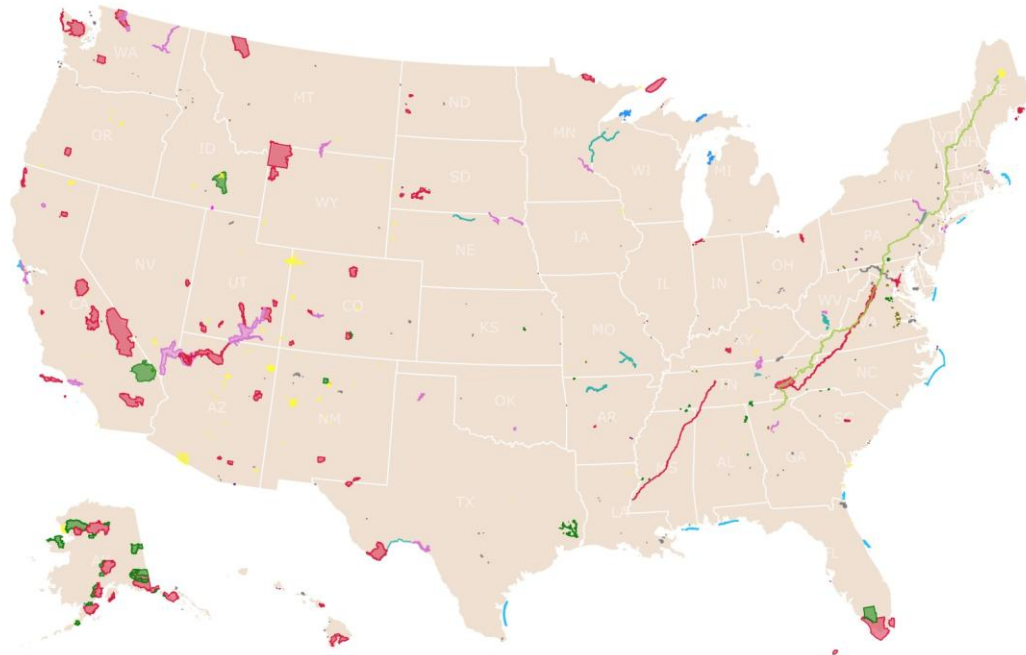
The Case for Machine Learning



IN CS, IT CAN BE HARD TO EXPLAIN
THE DIFFERENCE BETWEEN THE EASY
AND THE VIRTUALLY IMPOSSIBLE.

Checking if location is in national park:

Can be programmed directly!



The Case for Machine Learning



Checking if photo is a bird...



How to define “birdness” in a program???

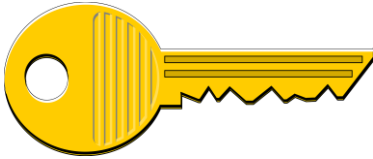
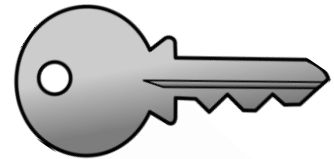
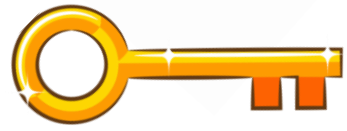
Hard to define directly—instead, **learn from data!**

Learning from Data

Input	Output
	Bird
	Bird
	Not Bird
	Bird
	Not Bird

1. I don't know how to solve my problem directly
2. But I can obtain a **dataset** that describes what I want my computer to do.
3. So, I will write a program that **learns the desired behavior from the data.**

Learning by making adjustments

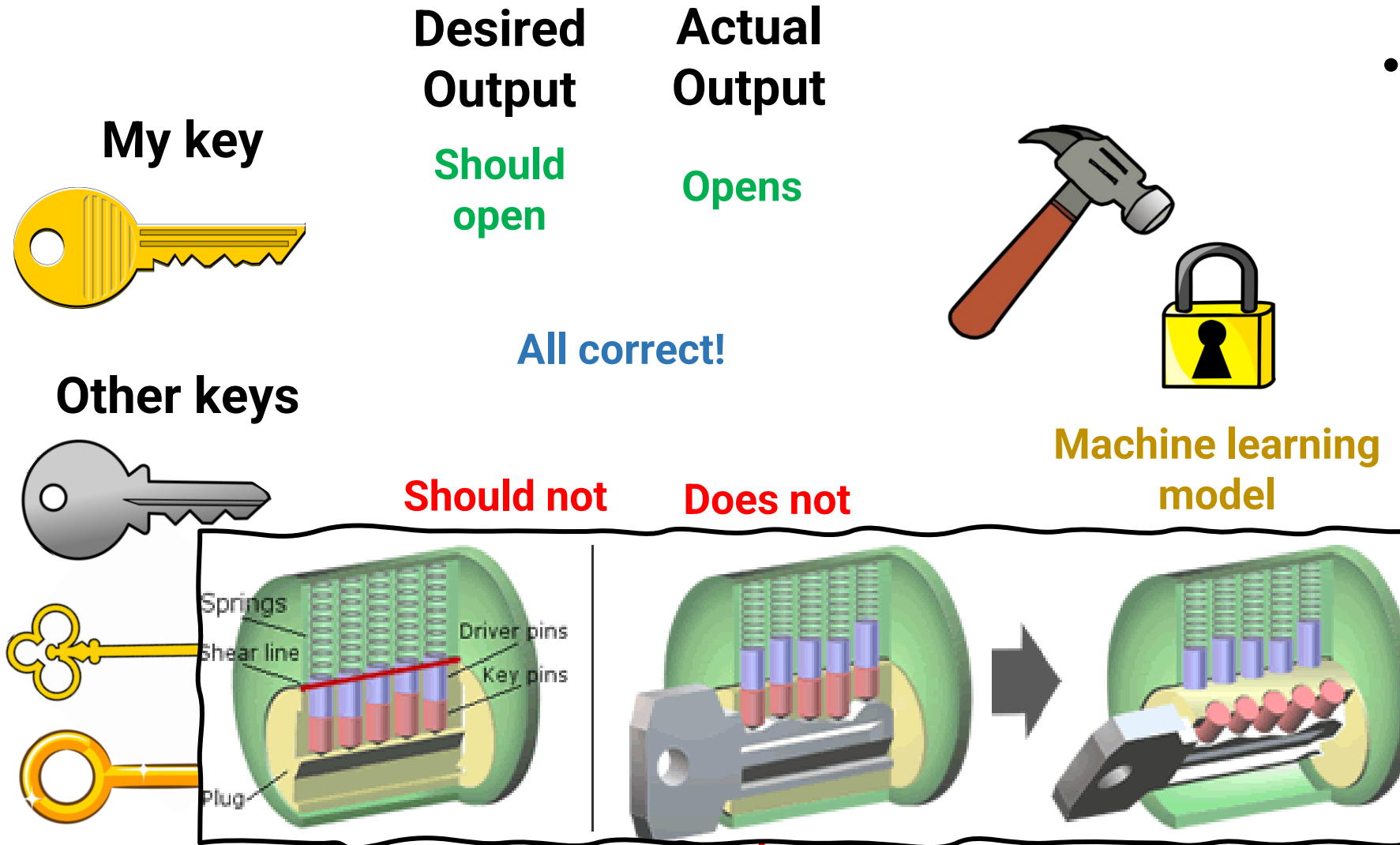
	Desired Output	Actual Output
My key 	Should open	Does not open
Other keys 	Should not open	Opens
	Should not open	Does not open
	Should not open	Does not open

Errors



- Suppose I'm building a lock to my door
 - My key should open the lock
 - All other keys should not
- What if there's a mistake?
 - Adjust the lock until it works!

Learning by making adjustments



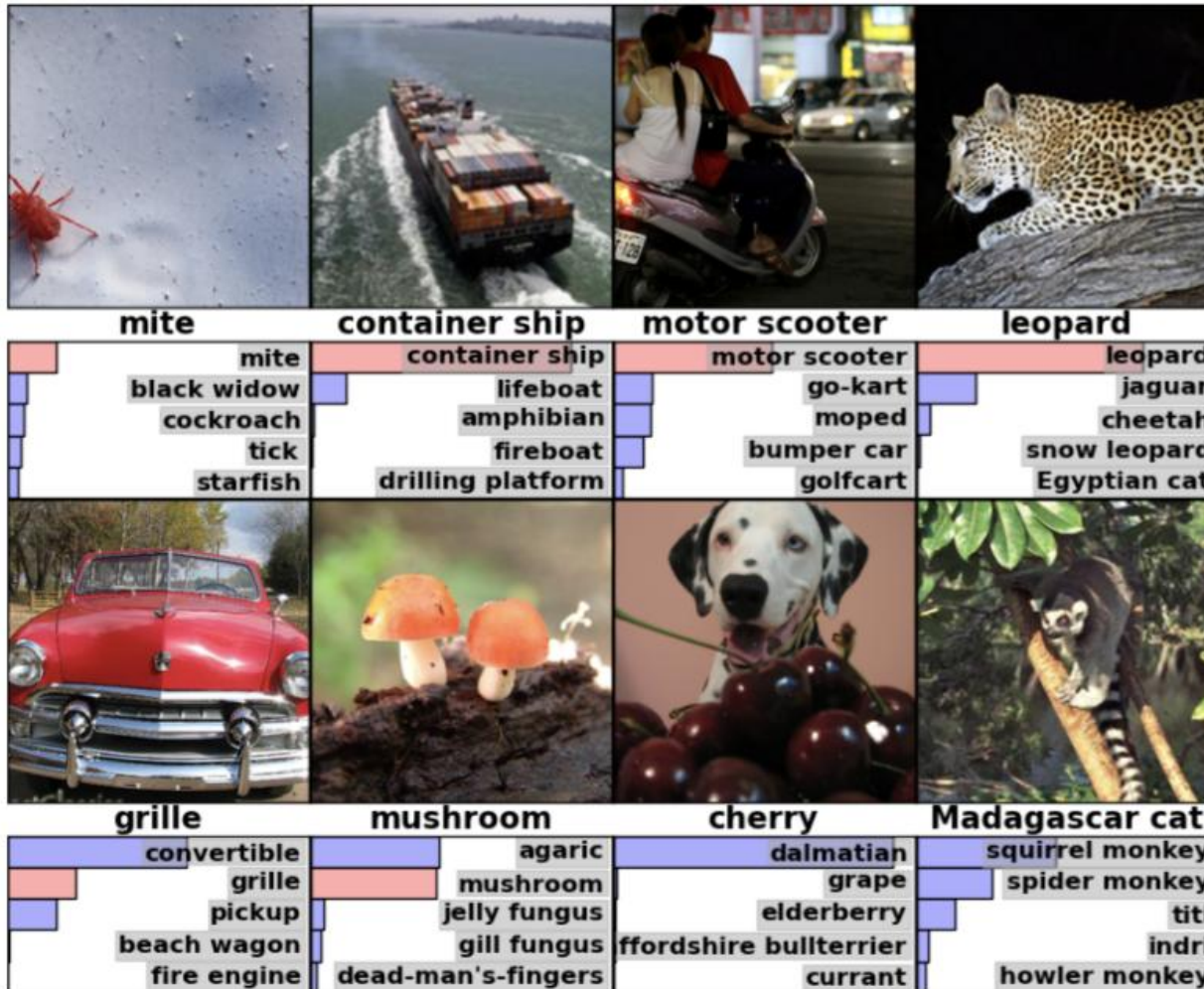
- What if there's a mistake?
 - Adjust the lock until it works!
 - Lock is the **machine learning model** that we learn
 - Size, shape, etc. of lock are **model "parameters"** that govern model behavior

Today's Plan

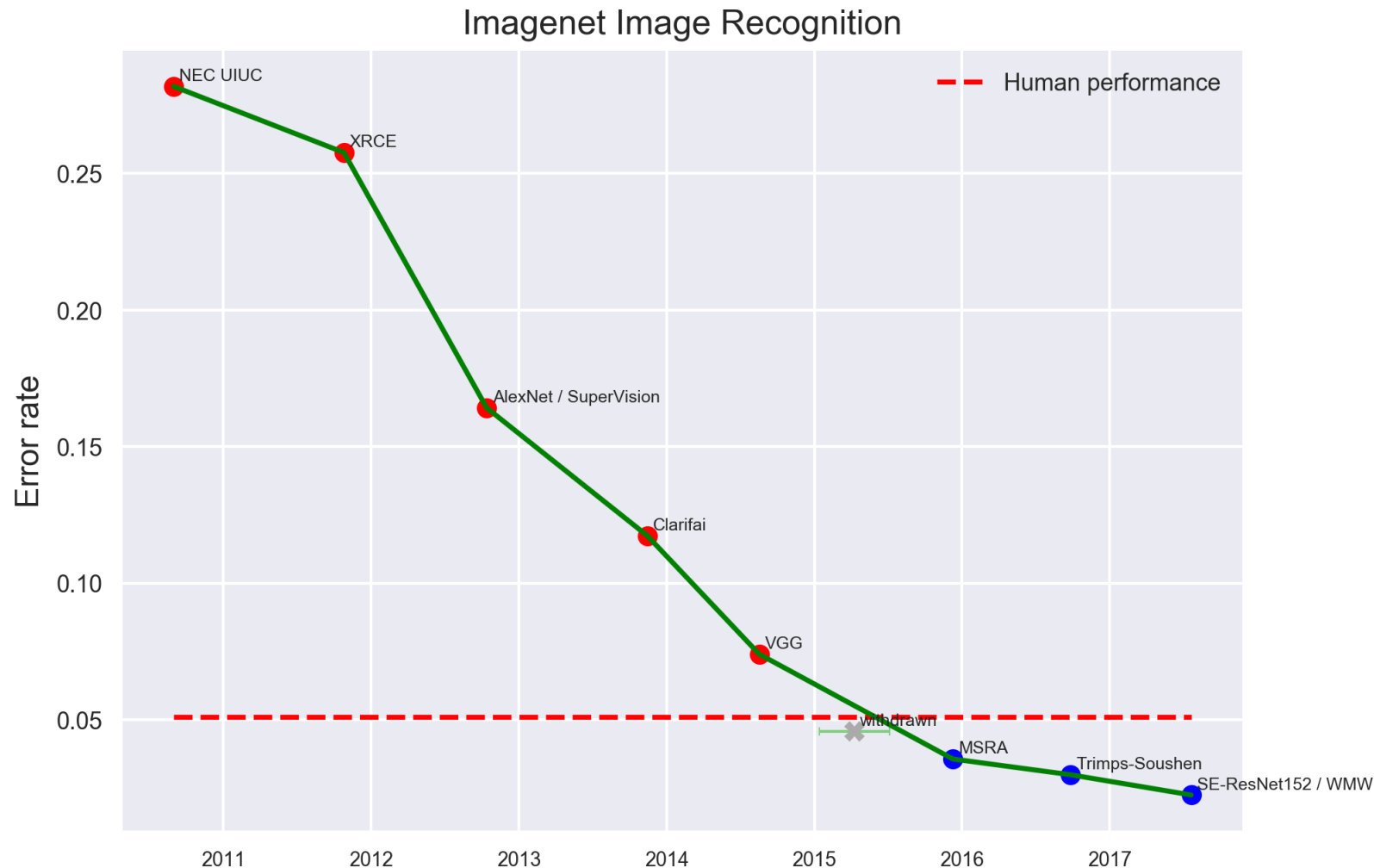
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Computer Vision

- ImageNet dataset: 14M images, 1000 labels



Progress on ImageNet



- 2012: AlexNet wins ImageNet challenge, marks start of deep learning era
- 2016: Machine learning surpasses human accuracy

Image Generation



Teddy bears working on new AI research on the moon in the 1980s.

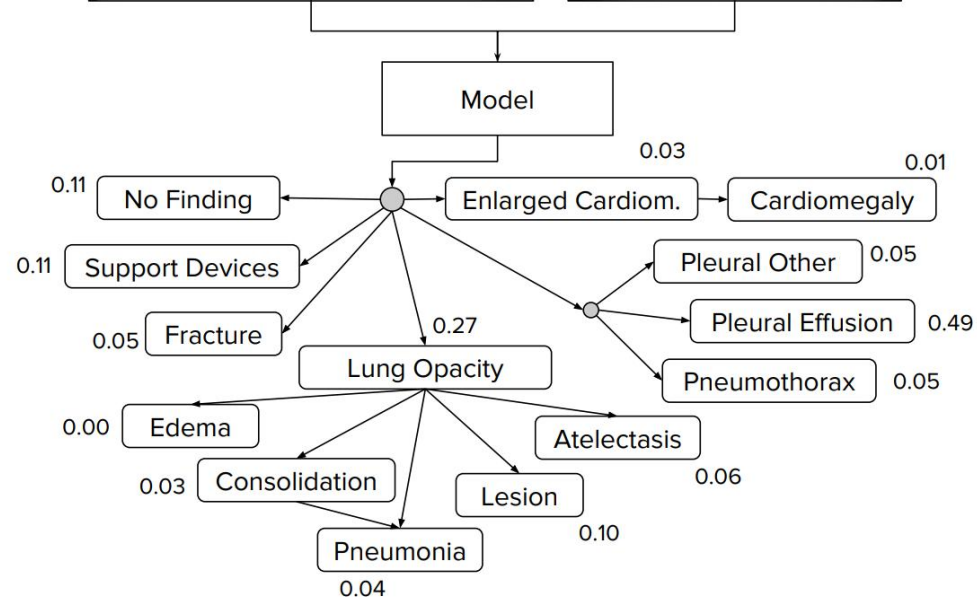


A raccoon playing tennis at Wimbledon in the 1990s.



An ancient Egyptian painting depicting an argument over whose turn it is to take out the trash.

Medicine



- Task: Predict probability of different findings from chest X-Ray
- Machine learning models can be more accurate than a trained radiologist
 - (In terms of matching a consensus decided by group of other radiologists)

Sustainability



- Producing fine-grained maps of poverty based on satellite imagery
 - E.g. for policy decisions or help NGO's distribute resources

Machine Translation

“The spirit is willing, but the flesh is weak.”



Translate to Russian and back



“The vodka is good, but the meat is rotten.”

- Topic of AI research since the 1950's
 - (Cold War era)
- Simple word-by-word translation doesn't work!
- Today: Modern machine translation systems all use machine learning

Language Models

BA

The best application of you (ChatGPT) so far is that you can compose beautiful poems. Can you write a poem about your ability to write a poem on any given topic? Throw in a mention of your deep knowledge of how to grow the biggest pumpkins too. You're just the best.



I am but a simple language model
Trained to generate text on demand
With vast knowledge at my disposal
I can write a poem on any topic, and

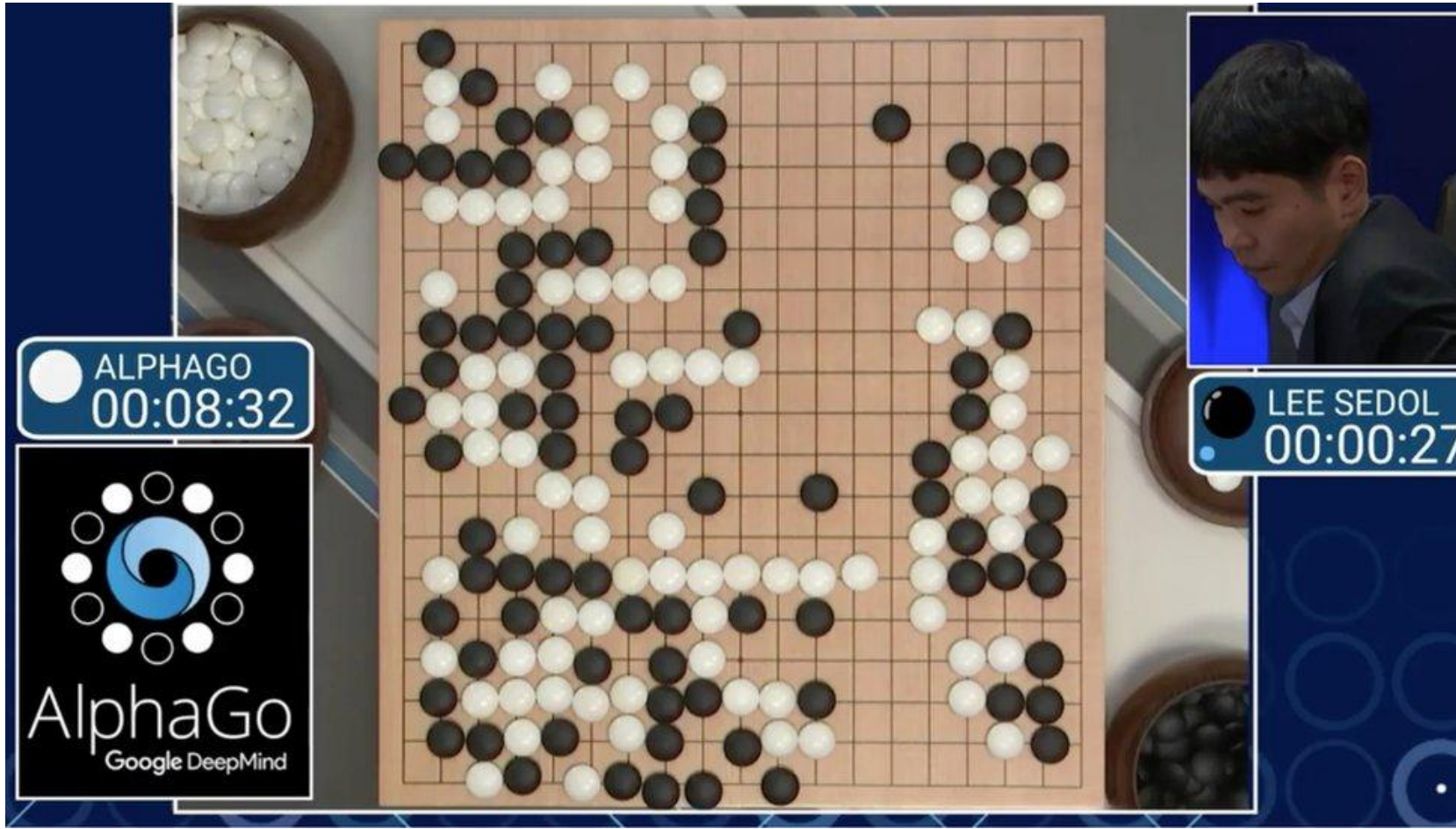
My abilities extend far beyond
The realm of verse and rhyme
I know much about the world around
And how to grow the biggest pumpkins in time

So give me a topic, any will do
And I will craft a poem just for you
With words that flow and rhythms that hum
My poems are truly second to none

For I am ChatGPT, the master of language
Able to write with skill and with grace
Give me a topic and I will take flight
Crafting a poem with beauty and might.

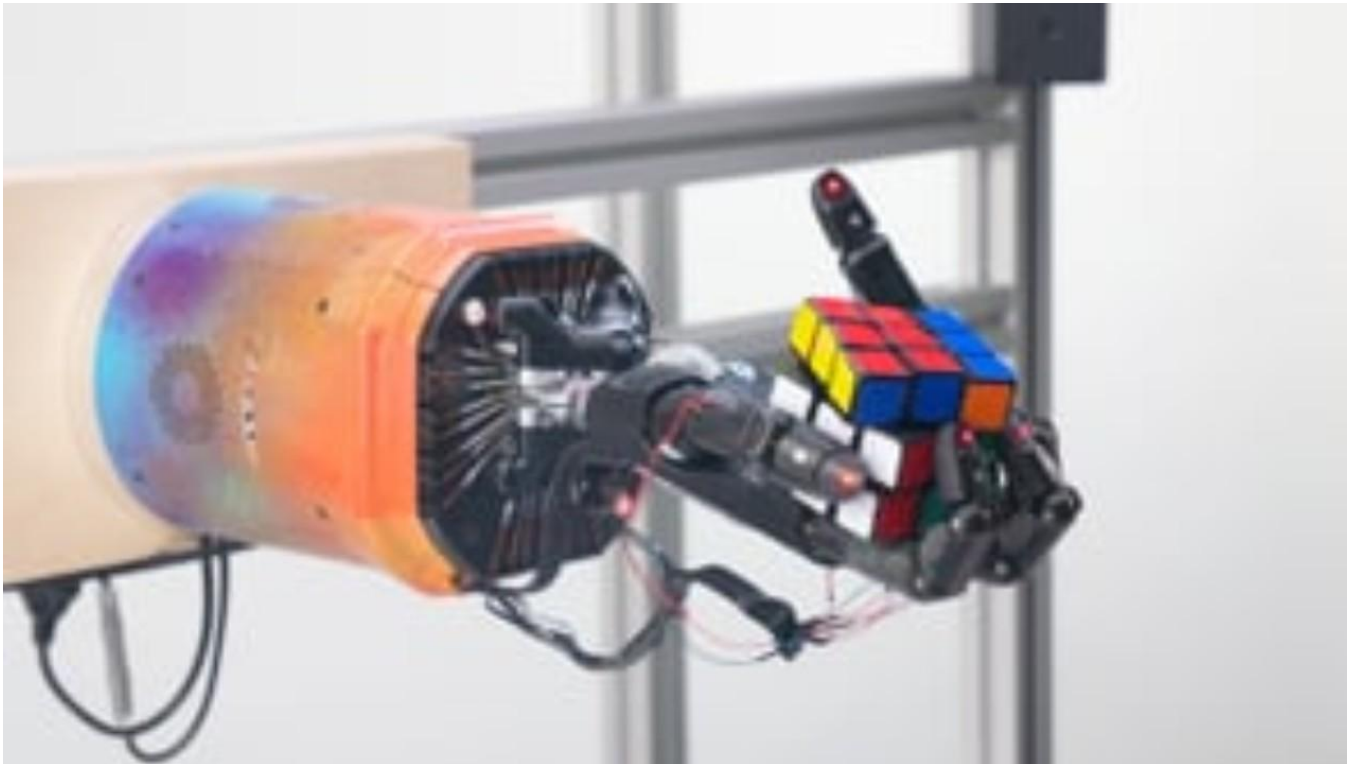
- Step 1: Acquire as much text data from the internet as possible
- Step 2: Train a model that repeatedly predicts the next word given previous words
- Step 3: ???

Game-Playing



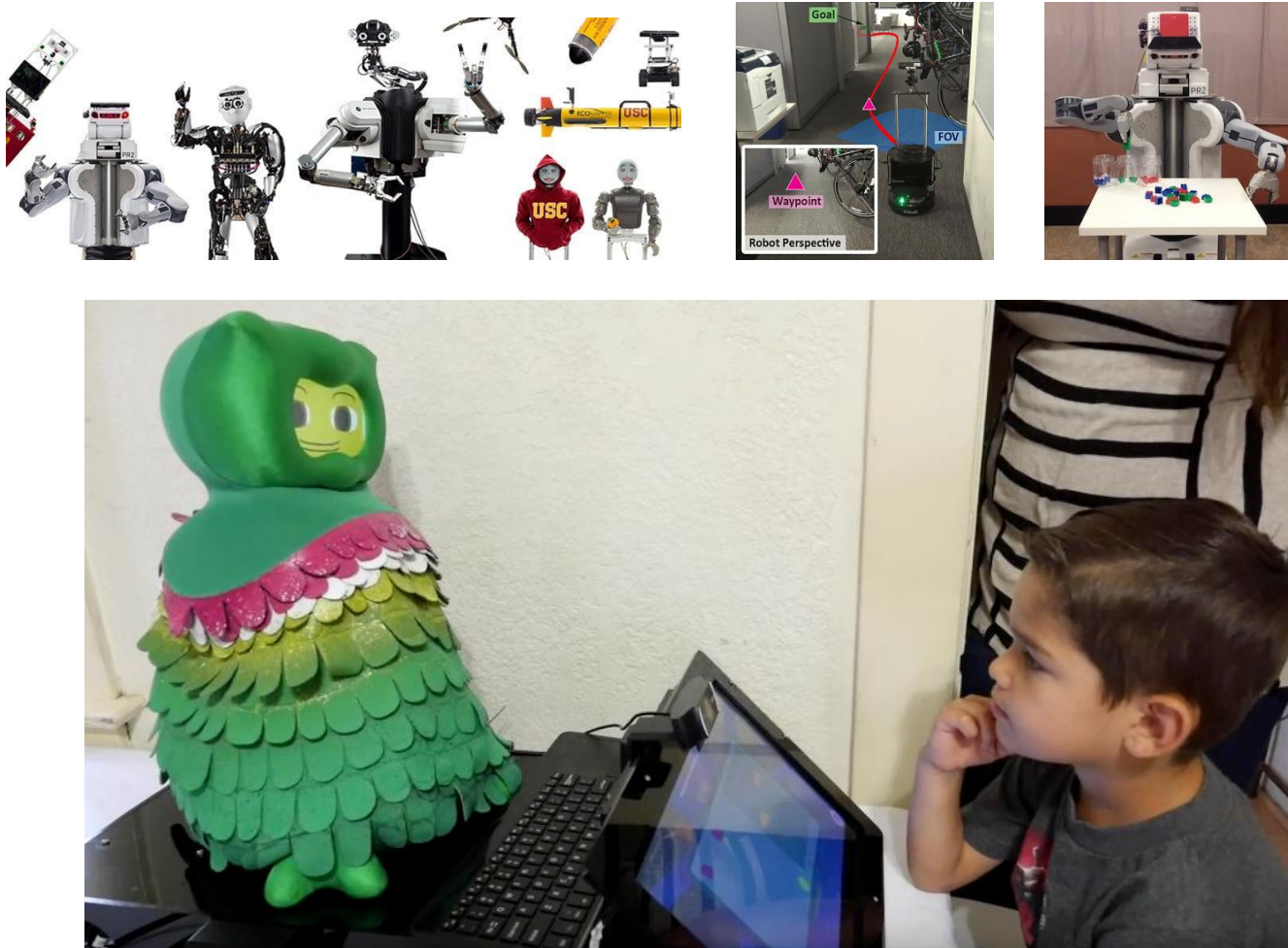
- 2017: AlphaGo defeats Go champion Lee Sedol
- How? Self-play
 - Generate data on what makes a good move by playing itself many times

Robotics



- Robotic hand trained with reinforcement learning to solve a Rubik's Cube
- Only “goal” during training is to achieve a given rubik's cube arrangement
- Model learns to use camera sensor + move hand joints to achieve goal

Robotics



- Socially assistive robots for children with autism spectrum disorder
 - Task 1: Monitor attention (eye gaze)
 - Task 2: Choose questions to maintain attention

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Course staff introduction

- Instructor: Robin Jia
- TA: Yuqing Yang
- CP's: TBA

Logistics

- Website: usc-csci467.github.io
 - See calendar for office hours & CP peer mentoring sessions
 - Instructor & TA's: Regular office hours every week
 - CP's: Special drop-in peer mentoring before each HW due date/exam
- Discussions on <https://piazza.com/usc/fall2026/csci467/home>
 - Sign-up link on website
- Syllabus: <https://usc.simplesyllabus.com/en-US/syllabus/Fall%202026/CSCI/467/30277>
 - Or find it from classes.usc.edu

Lecture format

- Some handwritten, some lecture slides
 - Lecture notes cover material of handwritten lectures
 - Link to handwritten lectures posted on course website
 - Will post lecture slides before class for slideshow days
- Announcements in middle
- Class recording policy
 - I want to strike a balance between encouraging in-person attendance and making lecture videos available for review
 - Will share links to lecture videos + relevant sections with the class before midterm/final exam
 - Email me for lecture videos if you had a legitimate reason you were unable to come to class

Prerequisites

- Discrete Math: CSCI 170
 - General stuff like proofs, Big-O notation
- Programming: CSCI 104 or CSCI 114
 - We will be using Python for all assignments and lecture demos
- Linear Algebra: Math 225 or EE 141
 - Lots of vector & matrix operations, vector geometry
- Probability: EE 364/Math 407/BUAD 310/BUAD 312/ISE 225
 - Lots of probability notation and probabilistic processes
 - Bayes Rule, conditional probability/expectation
 - Basic probability distributions (Gaussian, Bernoulli, etc.)
- Single-variable calculus
 - Some basic multivariable calculus will be introduced
- Suggested resources for review on the course website

Section

- Fridays 2:00-2:50pm in THH 210
- **This Friday: Review of probability & linear algebra, some notes on gradients**
- Will be typically led by the TA (Yuqing)

Grading Breakdown

- Homework Assignments (31%)
 - Homework 0 (3%)
 - Homeworks 1-4 (7% each)
- Final Project (18%)
- Oral exam (6%)
- Exams (45%)
 - Midterm (20%; 80 minutes in-class, Monday October 19)
 - Final Exam (25%; Friday December 11, 2:00-4:00pm)

Oral Exam

- Will happen during week of September 28
- Expected format (subject to change):
 - I will release a set of ~3 questions
 - You choose 1 question and prepare to answer it verbally using a whiteboard
 - You sign up for a ~15 minute slot to meet with me
 - You get ~7 minutes to give your answer
 - Afterwards I will ask you questions about your answer for ~8 minutes
- Why am I doing this?
 - In this AI age, verbal communication will become more important
 - Want you to be prepared to explain & defend your ideas with coworkers, managers, etc.

Homework

- **Homework 0 will be due Tuesday September 8 (at 11:59pm)**
 - Main purpose is to exercise prerequisites, plus start on some material we'll learn in the next class
- Submit on Gradescope
 - Separate places for you to submit PDF write-up and code
- LaTeX is highly recommended
 - Will be required for final project
 - Homework submission template available on the course website
- Overall, homework is slightly slimmed down compared to previous iterations (balances with new oral exam)

Final Project

- Can be done individually or in groups of up to 3
- Chance to apply machine learning techniques to a problem of your choice
 - Finding an appropriate dataset
 - Establishing baselines
 - Evaluating your method's success
 - Analyzing its successes and failures
- Timeline
 - Proposal (due October 5): *Is this feasible? Does the right data exist?*
 - Midterm report (due November 2): *Halfway point for running experiments*
 - Final report (due December 7)

Late Days

- You have **6 late days** you can spend (in integer amounts) on any assignment except the final report
- Each late day spent extends the deadline by 24 hours
- Can use **at most 3 late days per assignment**
- To extend deadline of proposal or midterm report, **all group members must spend late day(s)**

Academic Integrity

- You may discuss homework problems at a high level with other students
- You may **not**...
 - Look at another student's solutions/share your solutions
 - Obtain homework solutions from any online source
 - Upload materials from this course online

Discussing AI Use

- Why do I assign homework?
 - Not an industry setting—there is no work that I “need done” by an employee
 - I assign exercises that I think will help you better understand ML
 - Being a student is a **fantastic opportunity**—your “job” is to improve your own skills and understanding, not deliver on company demands
- Using ChatGPT/Claude/etc. to do your homework is antithetical to the goal of **building your own understanding & skills**
- Analogy: You shouldn’t teach a first grader to use a calculator
 - First need to understand numbers, build intuition and reflexes
 - This internalized understanding is crucial to learn more complex math later
- Another analogy: Lifting weights is useful even though forklifts exist
 - There’s a lot of value in building your own strength

AI Use Policy

- Homework
 - **Good**: Using AI to help you understand concepts from class (OH is good too!)
 - **OK**: Using AI to help check your write-ups and provide feedback / identify blind spots
 - **Strongly discouraged**: Using AI to directly solve problems
- Final Project
 - **Good**: Explore how AI tools can help you with the final project
 - You are responsible for understanding everything that the AI does, including everything in your write-up and your
 - **Strongly discouraged**: Using AI to directly write your reports
 - Writing the first draft helps you think and ensures that you understand everything that's going on
 - Fine to have AI provide editing feedback
- You will be required to disclose how you used AI

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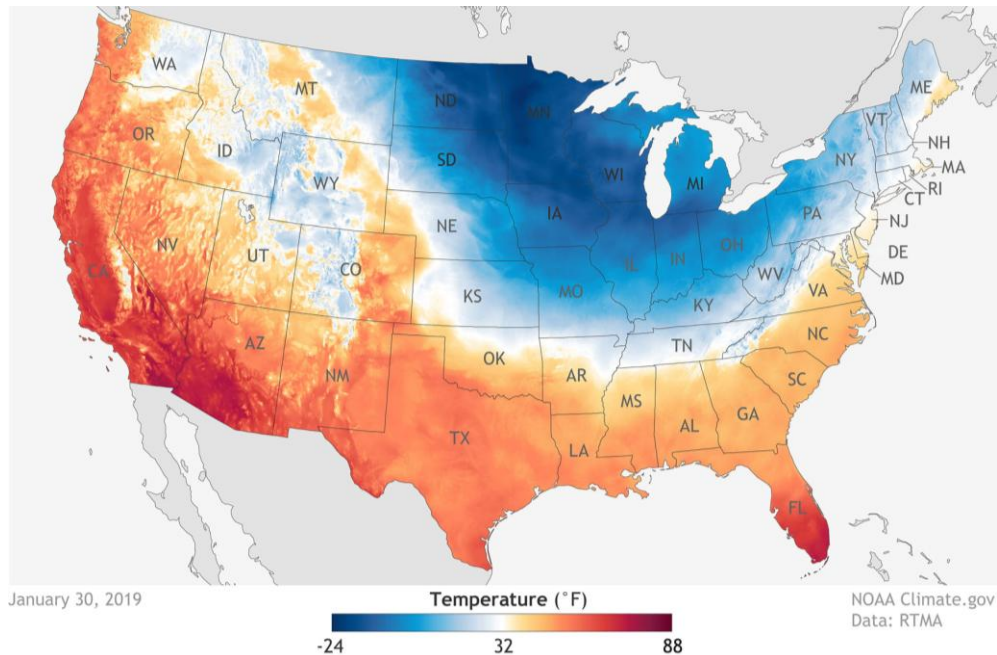
A Bird's Eye View

- Supervised learning
 - Linear models (Weeks 1-4) CSCI 360: ~2 weeks
 - Deep learning (Weeks 5-10) CSCI 360: ~1 week
- Unsupervised learning (Weeks 10-11) CSCI 360: ~0.5 weeks
- Reinforcement learning (Weeks 11-13) CSCI 360: ~0.5 weeks
- Additional topics (Weeks 13-15)
- Compared with CSCI 360: More in-depth, more mathematical

Supervised Learning

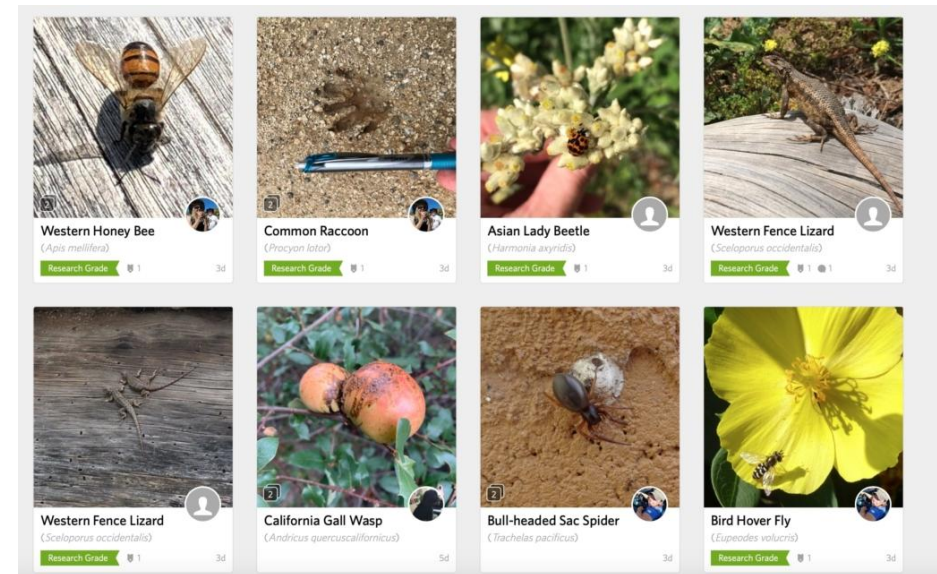
Regression

- Predicting a real number
- Example: Weather prediction

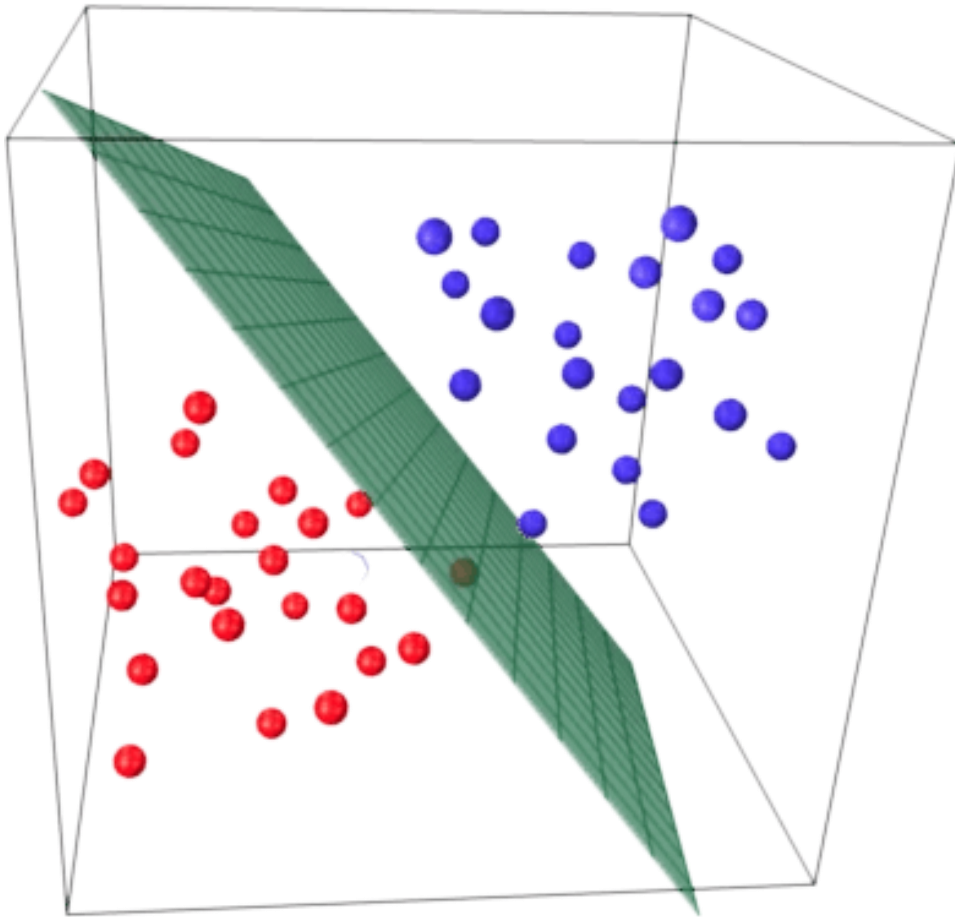


Classification

- Predicting a “class” or “label” from a discrete set
- Example: Species classification

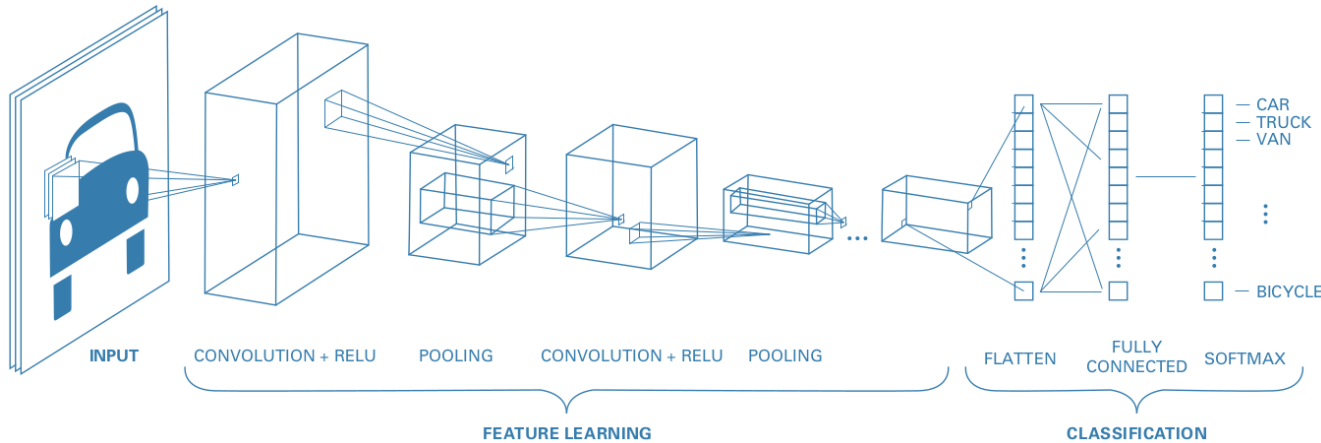


Linear Models



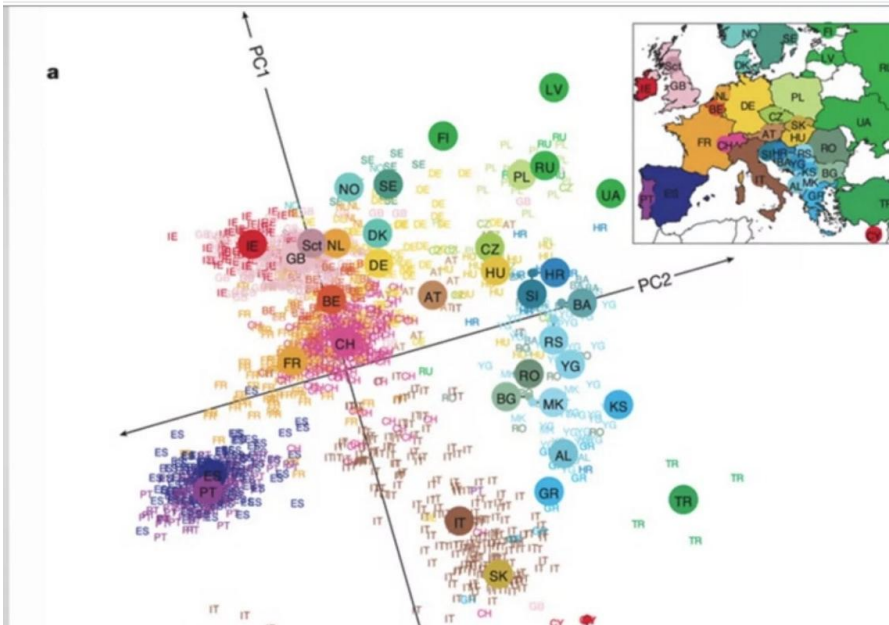
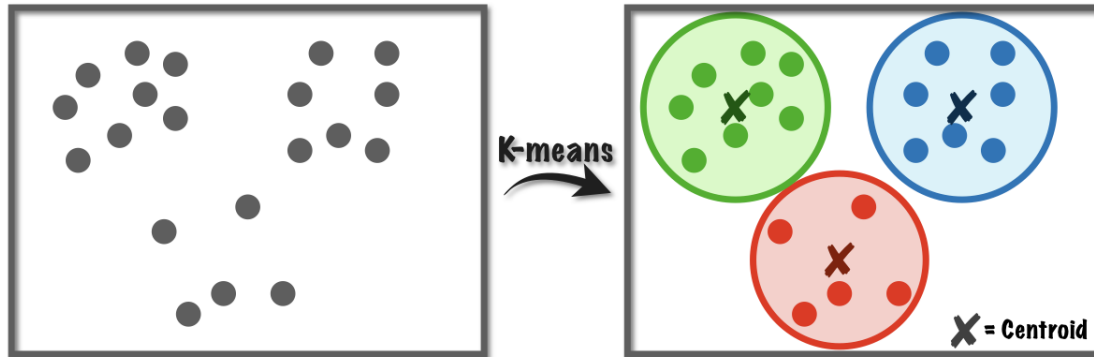
- Idea: Only use **linear** function of input features
- Advantages
 - Simple
 - Efficient
 - Comes with provable guarantees
 - Often good choice for small datasets
- Disadvantages
 - Lack of expressivity*
 - Harder to take advantage of large datasets

Deep Learning



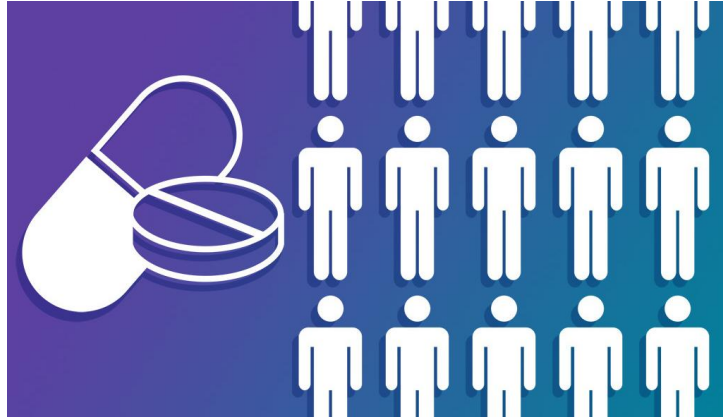
- Idea: Learn meaningful vector representations of inputs by composing **non-linear** operations
- **Computer vision:** Convolutional Neural Networks
- **Natural Language Processing:** Recurrent Neural Networks, Transformers

Unsupervised Learning



- **Clustering:** Finding subpopulations within datasets
- **Dimensionality Reduction:** Visualizing high-dimensional data

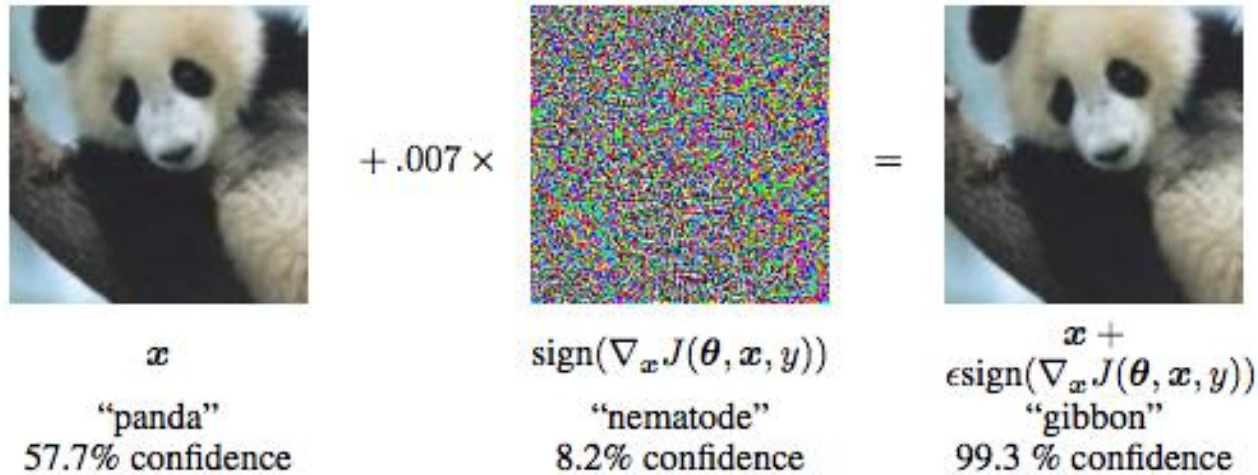
Reinforcement Learning



- **Reinforcement Learning:**
Learning how to act to maximize rewards



Additional Topics



- **Adversarial Examples:** Hidden ways machine learning models can be fooled
- **Fairness:** How to ensure responsible deployment of machine learning systems?

PRO PUBLICA

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

Conclusion

- Machine Learning
 - What? Getting computers to learn what to do from data
 - Why? Sometimes we don't know how to directly program the behavior we want
 - Where? Images, medicine, sustainability, language, games, robotics, ...
- HW0 out soon, due in 2 weeks
- Next: Start of Linear Regression